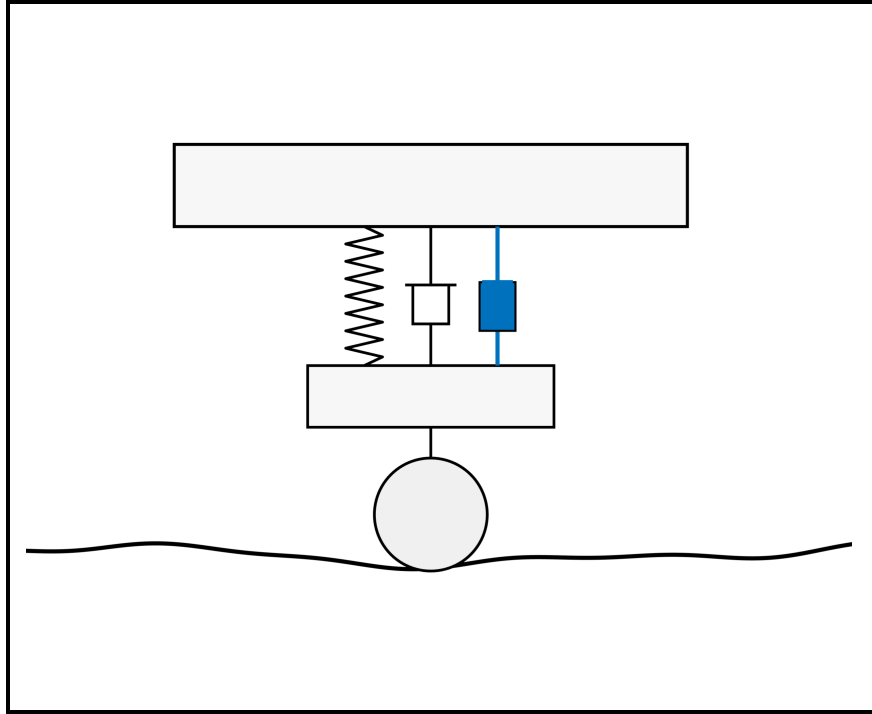


Development and Robustness Evaluation of a Sensor-Aware Gain-Scheduled LQR Active Suspension Controller in MATLAB

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Abstract

This study investigates the development of an active suspension control architecture for a quarter-car vehicle model, with emphasis on the trade-off between ride comfort, suspension travel, tire-road interaction, and actuator effort.

A passive suspension was first used as the baseline model. A fixed Linear Quadratic Regulator (LQR) was then introduced, demonstrating the benefit of active control while retaining a single control compromise across different disturbance conditions. To vary these priorities, a bank of Balanced, Comfort, Low Force, and Protection LQR controllers was developed. Model 3A manually scheduled these controllers using known road position, whereas Model 3B replaced this prior road knowledge with a hierarchical response-based scheduler using measured body acceleration, wheel acceleration, suspension travel, and relative suspension velocity.

The final Model 3B implementation included a ± 750 N actuator-force limit, a 50,000 N/s force-rate limit, 500 Hz sensor sampling, 4 ms transport delay, and measurement-noise effects. On the common benchmark road, Model 3B reduced peak body acceleration, peak suspension travel, and peak tire deflection by approximately 12.4%, 8.3%, and 8.0%, respectively, relative to the fixed Balanced LQR controller.

Robustness tests involving different bump heights, vehicle speeds, unseen mixed-road profiles, sensor delay, and measurement noise were performed without retuning the controller or scheduler parameters. The results support response-based gain-scheduled LQR control as a feasible proof-of-concept approach for adapting suspension priorities according to the measured vehicle response rather than prior knowledge of the road profile.

1 Quarter-Car Model and Performance Metrics

A two-degree-of-freedom quarter-car model was used to investigate the vertical dynamics of the suspension system. The model represents one quarter of the vehicle and consists of a sprung mass, corresponding to the supported vehicle body, and an unsprung mass, representing the wheel and associated suspension components. The two masses are connected through the suspension spring and damper, while the tire is represented by a linear stiffness connecting the unsprung mass to the road surface.

An active actuator is placed between the sprung and unsprung masses and generates the control force F_a . The road displacement z_r acts as the external disturbance input to the system. The vertical displacements of the sprung and unsprung masses are denoted by z_s and z_u , respectively. A schematic representation of the quarter-car model used throughout this study is shown in Fig. 1.

The model considers only the vertical motion of the vehicle and therefore neglects pitch, roll, and lateral dynamics. Both the suspension spring and the tire are assumed to behave linearly, while the damping force is proportional to

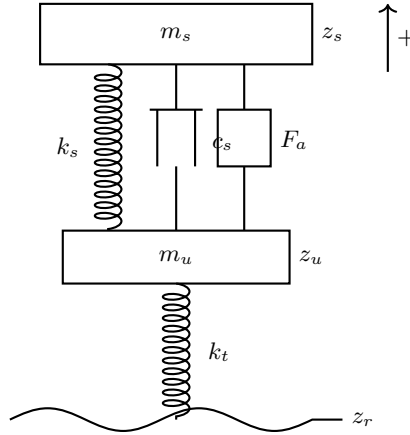


Figure 1: Two-degree-of-freedom active quarter-car suspension model.

the relative vertical velocity between the sprung and unsprung masses. Tire damping is neglected.

The displacement coordinates are defined relative to the static equilibrium position of the system. Consequently, gravitational forces do not explicitly appear in the dynamic equations. This formulation allows the analysis to focus on the dynamic response of the suspension to road disturbances and actuator forces.

1.1 Equations of Motion and Static Equilibrium

The nominal parameter set was selected to represent a typical passenger-car quarter-car model and was based on values reported in the suspension literature. For example, Soong et al. [1] reported passenger-car values of $m_s = 317.5$ kg, $m_u = 45.4$ kg, $k_s = 22\,000$ N/m, $c_s = 1500$ N s/m, and $k_t = 192\,000$ N/m. The values adopted in the present study were slightly rounded to obtain a simple nominal parameter set while preserving the same physical scale. Comparable quarter-car parameter ranges are also reported in other vehicle vertical-dynamics studies [2].

Table 1: Nominal quarter-car model parameters.

Parameter	Symbol	Value	Unit
Sprung mass	m_s	300	kg
Unsprung mass	m_u	40	kg
Suspension stiffness	k_s	22000	N/m
Suspension damping coefficient	c_s	1500	N s/m
Tire stiffness	k_t	200000	N/m
Gravitational acceleration	g	9.81	m/s ²

Before the dynamic equations are derived, the static deformation of the suspension and tire springs must be considered. When the vehicle is at rest on a flat road, both springs are already compressed under the weight of the vehicle. Therefore, the equilibrium configuration is not the undeformed configuration of the springs.

Let δ_{s0} denote the static compression of the suspension spring and δ_{t0} the static compression of the tire. Static force equilibrium of the sprung mass gives

$$k_s \delta_{s0} = m_s g, \quad (1)$$

and therefore

$$\delta_{s0} = \frac{m_s g}{k_s}. \quad (2)$$

For the unsprung mass, the tire supports both the unsprung mass and the load transmitted through the suspension spring. Hence,

$$k_t \delta_{t0} = k_s \delta_{s0} + m_u g. \quad (3)$$

Using $k_s \delta_{s0} = m_s g$,

$$k_t \delta_{t0} = (m_s + m_u) g, \quad (4)$$

which gives

$$\delta_{t0} = \frac{(m_s + m_u) g}{k_t}. \quad (5)$$

The dynamic coordinates z_s and z_u are measured from these static equilibrium positions, with the positive vertical direction defined upward. Consequently, the instantaneous suspension spring compression can be written as

$$\delta_s = \delta_{s0} - (z_s - z_u), \quad (6)$$

while the tire compression is

$$\delta_t = \delta_{t0} - (z_u - z_r). \quad (7)$$

Applying Newton's second law to the sprung mass gives

$$m_s \ddot{z}_s = k_s [\delta_{s0} - (z_s - z_u)] - c_s (\dot{z}_s - \dot{z}_u) + F_a - m_s g. \quad (8)$$

Using the static equilibrium relation

$$k_s \delta_{s0} = m_s g, \quad (9)$$

the constant gravitational and static spring-force terms cancel, resulting in

$$\boxed{m_s \ddot{z}_s = -k_s (z_s - z_u) - c_s (\dot{z}_s - \dot{z}_u) + F_a}. \quad (10)$$

Similarly, Newton's second law for the unsprung mass gives

$$m_u \ddot{z}_u = k_t [\delta_{t0} - (z_u - z_r)] - k_s [\delta_{s0} - (z_s - z_u)] + c_s (\dot{z}_s - \dot{z}_u) - F_a - m_u g. \quad (11)$$

The constant terms in this equation are

$$k_t \delta_{t0} - k_s \delta_{s0} - m_u g. \quad (12)$$

From the static equilibrium conditions,

$$k_t \delta_{t0} = k_s \delta_{s0} + m_u g, \quad (13)$$

and therefore these terms cancel exactly. The dynamic equation of the unsprung mass becomes

$$\boxed{m_u \ddot{z}_u = k_s (z_s - z_u) + c_s (\dot{z}_s - \dot{z}_u) - k_t (z_u - z_r) - F_a}. \quad (14)$$

Thus, gravity does not disappear as an assumption. Instead, its effect is already contained in the static deformation of the suspension and tire springs. By defining z_s and z_u relative to the static equilibrium configuration, the constant gravitational terms are removed from the equations, leaving only the dynamic response to road disturbance and actuator force.

1.2 State-Space Representation

For the development of the active suspension controllers, the equations of motion were expressed in state-space form. Starting from the dynamic equations derived in the previous section,

$$m_s \ddot{z}_s = -k_s (z_s - z_u) - c_s (\dot{z}_s - \dot{z}_u) + F_a, \quad (15)$$

and

$$m_u \ddot{z}_u = k_s (z_s - z_u) + c_s (\dot{z}_s - \dot{z}_u) - k_t (z_u - z_r) - F_a, \quad (16)$$

the acceleration terms can be written explicitly as

$$\ddot{z}_s = -\frac{k_s}{m_s} z_s - \frac{c_s}{m_s} \dot{z}_s + \frac{k_s}{m_s} z_u + \frac{c_s}{m_s} \dot{z}_u + \frac{1}{m_s} F_a, \quad (17)$$

and

$$\ddot{z}_u = \frac{k_s}{m_u} z_s + \frac{c_s}{m_u} \dot{z}_s - \frac{k_s + k_t}{m_u} z_u - \frac{c_s}{m_u} \dot{z}_u - \frac{1}{m_u} F_a + \frac{k_t}{m_u} z_r. \quad (18)$$

The state vector was selected as

$$\mathbf{x} = \begin{bmatrix} z_s \\ \dot{z}_s \\ z_u \\ \dot{z}_u \end{bmatrix}. \quad (19)$$

The system can then be written in the form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}F_a + \mathbf{E}z_r, \quad (20)$$

where $\mathbf{A}$ describes the vehicle dynamics, $\mathbf{B}$ represents the influence of the actuator force, and $\mathbf{E}$ represents the road disturbance input.

The corresponding matrices are

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_s}{m_s} & -\frac{c_s}{m_s} & \frac{k_s}{m_s} & \frac{c_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{c_s}{m_u} & -\frac{k_s + k_t}{m_u} & -\frac{c_s}{m_u} \end{bmatrix}, \quad (21)$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{m_s} \\ 0 \\ -\frac{1}{m_u} \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{k_t}{m_u} \end{bmatrix}. \quad (22)$$

The opposite signs in $\mathbf{B}$ arise because the actuator is located between the sprung and unsprung masses and therefore applies equal and opposite forces to them. The road displacement enters the system through the tire deformation and directly affects the unsprung-mass acceleration.

This state-space formulation is used as the common plant model throughout the study. The passive, fixed-LQR, and gain-scheduled models therefore share the same vehicle dynamics, while only the control law used to determine F_a is changed.

1.3 Performance Metrics

The performance of the suspension system cannot be characterized by a single response quantity. In an active suspension system, improving one aspect of the vehicle response may require accepting a degradation in another. Therefore, four performance quantities were monitored throughout the study: body acceleration, suspension travel, tire deflection, and actuator force.

1.3.1 Ride Comfort

Ride comfort was evaluated using the vertical acceleration of the sprung mass,

$$a_s = \ddot{z}_s. \quad (23)$$

Two acceleration-based quantities were considered. The root-mean-square (RMS) acceleration was used to represent the overall vibration level during a simulation,

$$a_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T \ddot{z}_s^2(t) dt}, \quad (24)$$

while the peak body acceleration was defined as

$$a_{\text{peak}} = \max_t |\ddot{z}_s(t)|. \quad (25)$$

A lower RMS acceleration generally indicates improved ride comfort, whereas the peak acceleration is useful for evaluating short transient events such as bumps or severe road disturbances.

1.3.2 Suspension Travel

Suspension travel was defined as the relative displacement between the sprung and unsprung masses,

$$s(t) = z_s(t) - z_u(t). \quad (26)$$

The maximum absolute suspension travel was evaluated as

$$s_{\text{peak}} = \max_t |z_s(t) - z_u(t)|. \quad (27)$$

This quantity represents the relative motion required from the suspension. Excessive suspension travel may become undesirable because a real suspension system has a finite mechanical working range.

1.3.3 Tire Deflection

The tire deflection was defined as the displacement of the unsprung mass relative to the road input,

$$d_t(t) = z_u(t) - z_r(t). \quad (28)$$

Its maximum absolute value was calculated as

$$d_{t,\text{peak}} = \max_t |z_u(t) - z_r(t)|. \quad (29)$$

Within the linear tire model used in this study, tire deflection is related to the variation of the vertical tire force through

$$\Delta F_t = k_t(z_u - z_r). \quad (30)$$

Therefore, tire deflection was used as an indicator of road-holding behavior, with smaller values generally corresponding to smaller variations in tire force.

1.3.4 Actuator Effort

The control effort was evaluated directly through the active suspension force F_a . The RMS actuator force was defined as

$$F_{\text{RMS}} = \sqrt{\frac{1}{T} \int_0^T F_a^2(t) dt}, \quad (31)$$

and the peak actuator force was calculated as

$$F_{\text{peak}} = \max_t |F_a(t)|. \quad (32)$$

The RMS value provides a measure of the sustained actuator demand, whereas the peak force represents the maximum instantaneous force requirement.

1.3.5 Interpretation of the Performance Metrics

The four quantities above represent competing suspension objectives. For example, reducing body acceleration may require greater actuator effort or larger suspension travel, while reducing actuator force may allow greater body motion. Consequently, lower values of a single metric do not necessarily imply superior overall suspension performance.

This distinction becomes particularly important later in the study, where different control objectives are used under different vehicle response conditions. The desired suspension behavior is therefore evaluated according to the trade-off among ride comfort, suspension travel, tire response, and actuator effort rather than by minimizing one metric independently over all operating conditions.

1.4 Road Input and Simulation Framework

The road displacement z_r acts as the external disturbance to the quarter-car model. Rather than defining the road profile only as a function of time, it is convenient to describe the road geometrically as a function of longitudinal position,

$$z_r = z_r(x), \quad (33)$$

where x denotes the distance travelled along the road.

For a vehicle travelling at a constant longitudinal speed v , the distance travelled is related to time through

$$x = vt. \quad (34)$$

Therefore, the road disturbance experienced by the suspension can be expressed in the time domain as

$$z_r(t) = z_r(vt). \quad (35)$$

This formulation allows the same physical road profile to be evaluated at different vehicle speeds without changing its spatial geometry. Changing the vehicle speed instead changes the rate at which the suspension encounters the road disturbances.

For example, a spatially sinusoidal road profile may be expressed as

$$z_r(x) = A_r \sin\left(\frac{2\pi x}{\lambda}\right), \quad (36)$$

where A_r is the road amplitude and λ is the spatial wavelength. Substituting $x = vt$ gives

$$z_r(t) = A_r \sin\left(\frac{2\pi v}{\lambda}t\right). \quad (37)$$

The corresponding excitation frequency experienced by the vehicle is therefore

$$f = \frac{v}{\lambda}. \quad (38)$$

This relationship is important because the dynamic response of the suspension depends not only on the shape of the road surface but also on the speed at which the vehicle travels over it. The same road feature can therefore produce substantially different vehicle responses at different speeds.

Since the displacement coordinates were defined relative to the static equilibrium configuration, the simulations were initialized using

$$\mathbf{x}(0) = [0 \quad 0 \quad 0 \quad 0]^T. \quad (39)$$

Thus, the vehicle is initially assumed to be at rest at its static equilibrium position before encountering the road disturbance.

A base simulation interval of

$$T_s = 0.002 \text{ s} \quad (40)$$

was used throughout the numerical studies, corresponding to a base frequency of

$$f_s = 500 \text{ Hz}. \quad (41)$$

This common simulation framework was maintained for the passive, fixed-LQR, and gain-scheduled suspension models so that differences in their responses could be attributed to the control strategy rather than changes in the underlying plant or numerical setup.

2 Model-1, Passive Suspension

The passive suspension was first investigated to establish a baseline for the active-control models developed later in the study. The purpose of this stage was not to optimize the passive suspension parameters, but to examine the behavior and limitations of a suspension whose dynamic characteristics are determined entirely by fixed mechanical components.

2.1 Passive System Formulation

In the passive configuration, no active force is applied between the sprung and unsprung masses. Therefore,

$$F_a = 0. \quad (42)$$

The state-space model introduced in the previous section consequently reduces to

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{E}z_r. \quad (43)$$

The equations of motion of the passive system are therefore

$$m_s \ddot{z}_s = -k_s(z_s - z_u) - c_s(\dot{z}_s - \dot{z}_u), \quad (44)$$

and

$$m_u \ddot{z}_u = k_s(z_s - z_u) + c_s(\dot{z}_s - \dot{z}_u) - k_t(z_u - z_r). \quad (45)$$

Thus, the response of the passive suspension is governed entirely by the suspension stiffness k_s , suspension damping coefficient c_s , tire stiffness k_t , and the properties of the road disturbance. Unlike an active suspension, the passive system cannot modify the suspension force independently in response to changing operating conditions.

The nominal mechanical parameters defined in Table 1 were initially retained to establish a baseline passive response. The nominal mechanical parameters were retained throughout the subsequent model comparisons so that differences in performance could be attributed primarily to the control strategy.

2.2 Common-Road Passive Baseline

To provide a consistent basis for comparison between the passive and active suspension models, a common composite road profile was used throughout the nominal model evaluations. The same road geometry, vehicle speed, mechanical parameters, and initial conditions are therefore retained when Models 1, 2, 3A, and 3B are compared later in the study.

The benchmark road has a total length of

$$L_{\text{road}} = 80 \text{ m}, \quad (46)$$

and the vehicle travels at a constant longitudinal speed of

$$v = 10 \text{ m/s.} \quad (47)$$

The resulting simulation duration is therefore

$$T = \frac{L_{\text{road}}}{v} = 8 \text{ s.} \quad (48)$$

Rather than representing a single road condition, the composite profile contains several consecutive disturbance types. This allows the same suspension model to be exposed to low-frequency body motion, medium-frequency excitation, high-frequency wheel activity, and a localized transient disturbance within a single simulation.

The road sections used in the benchmark are summarized in Table 2.

Table 2: Sections of the common composite benchmark road.

Position	Road condition	Main characteristic
0–16 m	Mild smooth	2 mm, long wavelength
16–32 m	Long-wave undulation	15 mm, long wavelength
32–44 m	Medium-frequency	10 mm periodic disturbance
44–54 m	High-frequency rough	5 mm, short wavelength
54–64 m	Bump section	30 mm localized bump
64–72 m	Wheel-hop region	High-frequency periodic excitation
72–80 m	Mild smooth	2 mm, long wavelength

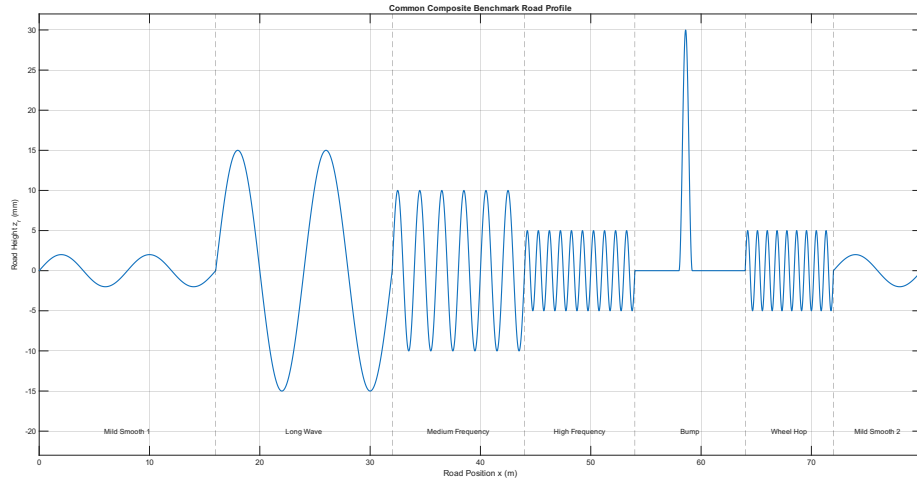


Figure 2: Common composite benchmark road used for the nominal comparison of the passive and active suspension models.

2.3 Passive Response Results

The passive suspension was simulated over the complete composite benchmark road using the nominal mechanical parameters. Since the road profile contains several distinct excitation conditions, the response was evaluated separately within each road segment in order to identify how the dominant performance characteristics changed with the disturbance type.

The segment-wise performance metrics are summarized in Table 3.

Table 3: Segment-wise performance of the passive suspension on the common composite benchmark road.

Road Segment	Start (m)	End (m)	Accel. RMS (m/s ²)	Accel. Peak (m/s ²)	Susp. Travel Peak (mm)	Tire Def. Peak (mm)
Mild Smooth 1	0	16	0.165	0.279	3.351	0.438
Long-Wave Undulation	16	32	1.269	2.102	25.252	3.302
Medium-Frequency	32	44	1.558	2.986	27.252	4.858
High-Frequency Rough	44	54	2.006	3.038	10.989	7.902
Bump Section	54	64	1.537	6.455	29.620	14.717
Wheel-Hop Region	64	72	2.227	3.491	11.394	10.042
Mild Smooth 2	72	80	0.450	2.362	8.199	9.101

The results demonstrate that the passive suspension response depends strongly on the character of the road excitation. The mild initial section produces relatively small body acceleration and suspension motion, whereas the long-wave and medium-frequency sections result in substantially larger suspension travel. The highest peak body acceleration occurs during the localized bump, while the wheel-hop region produces the largest RMS body acceleration among the individual road sections.

The dominant performance limitation therefore changes with the road condition. A passive suspension, however, cannot alter its mechanical characteristics once the parameters k_s and c_s have been selected. As a result, a parameter combination that provides an acceptable response under one type of excitation may become less suitable under another. This limitation motivates the introduction of active suspension control, where an additional actuator force can be used to modify the effective suspension response without changing the fixed mechanical components.

3 Model-2, Fixed Active LQR Control

Following the limitations observed in the passive suspension model, an active suspension force was introduced between the sprung and unsprung masses. The objective of the active controller is not to minimize a single response quantity independently, but to establish a controllable compromise between ride comfort, suspension motion, and actuator effort.

3.1 Control Objective and Cost Function

The active suspension force is generated according to a Linear Quadratic Regulator (LQR) control law. To retain a direct physical interpretation of the controller design, the cost function was defined in terms of measurable suspension-performance quantities rather than arbitrary state penalties.

The three quantities included in the control objective are the sprung mass acceleration,

$$\ddot{z}_s, \quad (49)$$

the suspension travel,

$$z_s - z_u, \quad (50)$$

and the actuator force,

$$F_a. \quad (51)$$

The corresponding cost function is written as

$$J = \int_0^\infty [w_a \ddot{z}_s^2 + w_s (z_s - z_u)^2 + w_F F_a^2] dt, \quad (52)$$

where w_a , w_s , and w_F represent the relative penalties assigned to body acceleration, suspension travel, and actuator force, respectively.

The individual terms represent competing control objectives. Increasing the acceleration penalty places greater emphasis on ride comfort, while increasing the suspension-travel penalty discourages large relative displacement between the sprung and unsprung masses. The actuator-force term penalizes excessive control effort and discourages the controller from relying on unnecessarily large actuator forces.

Rather than selecting the weighting coefficients directly, reference performance levels were introduced as

$$w_a = \frac{1}{a_{\text{ref}}^2}, \quad w_s = \frac{1}{s_{\text{ref}}^2}, \quad w_F = \frac{1}{F_{\text{ref}}^2}. \quad (53)$$

This scaling gives each term a comparable dimensionless form. A smaller reference value therefore corresponds to a stronger penalty on the associated quantity. For example, reducing a_{ref} increases the relative importance of body-acceleration reduction, whereas reducing F_{ref} increases the penalty on actuator usage.

The cost function consequently defines a specific compromise between the competing suspension objectives. Since the weighting coefficients remain fixed in the present model,

The resulting LQR controller still represents one **fixed active-suspension** compromise.

3.2 Generalized LQR Formulation

To implement the physical cost function within the LQR framework, the body acceleration and suspension travel were first expressed as functions of the state vector and actuator force.

From the sprung-mass equation of motion,

$$\ddot{z}_s = -\frac{k_s}{m_s}z_s - \frac{c_s}{m_s}\dot{z}_s + \frac{k_s}{m_s}z_u + \frac{c_s}{m_s}\dot{z}_u + \frac{1}{m_s}F_a, \quad (54)$$

the body acceleration can be written in compact form as

$$\ddot{z}_s = \mathbf{C}_a \mathbf{x} + D_a F_a, \quad (55)$$

where

$$\mathbf{C}_a = \begin{bmatrix} -\frac{k_s}{m_s} & -\frac{c_s}{m_s} & \frac{k_s}{m_s} & \frac{c_s}{m_s} \end{bmatrix}, \quad (56)$$

and

$$D_a = \frac{1}{m_s}. \quad (57)$$

Similarly, the suspension travel can be expressed as

$$z_s - z_u = \mathbf{C}_s \mathbf{x}, \quad (58)$$

with

$$\mathbf{C}_s = \begin{bmatrix} 1 & 0 & -1 & 0 \end{bmatrix}. \quad (59)$$

Substituting these expressions into the physical cost function gives

$$J = \int_0^\infty \left[w_a (\mathbf{C}_a \mathbf{x} + D_a F_a)^2 + w_s (\mathbf{C}_s \mathbf{x})^2 + w_F F_a^2 \right] dt. \quad (60)$$

Expanding the terms allows the cost function to be written in the generalized quadratic form

$$J = \int_0^\infty \left[\mathbf{x}^T \mathbf{Q} \mathbf{x} + 2\mathbf{x}^T \mathbf{N} F_a + R F_a^2 \right] dt, \quad (61)$$

where

$$\mathbf{Q} = w_a \mathbf{C}_a^T \mathbf{C}_a + w_s \mathbf{C}_s^T \mathbf{C}_s, \quad (62)$$

$$\mathbf{N} = w_a \mathbf{C}_a^T D_a, \quad (63)$$

and

$$R = w_a D_a^2 + w_F. \quad (64)$$

The cross-weighting term $\mathbf{N}$ appears because the sprung-mass acceleration depends directly on both the state vector and the actuator force. Therefore, penalizing body acceleration naturally produces a coupling term between the states and the control input in the quadratic cost function.

The optimal state-feedback control law is then written as

$$F_a = -\mathbf{K}\mathbf{x}, \quad (65)$$

where the gain matrix $\mathbf{K}$ is obtained by solving the generalized continuous-time LQR problem associated with $\mathbf{A}$, $\mathbf{B}$, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{N}$.

In the present controller, tire deflection is retained as a performance metric for evaluating road-holding behavior, but it is not directly included in the LQR cost function. The fixed controller is therefore designed primarily around the trade-off between body acceleration, suspension travel, and actuator effort.

3.3 Balanced Fixed-LQR Controller

For the fixed-LQR model, the body-acceleration and actuator-force reference values were retained at

$$a_{\text{ref}} = 2.0 \text{ m/s}^2, \quad F_{\text{ref}} = 500 \text{ N}, \quad (66)$$

while the suspension-travel reference was selected through a parameter sweep. Based on this study, the value

$$s_{\text{ref}} = 0.020 \text{ m} \quad (67)$$

was adopted for the fixed-LQR controller. These values correspond to the weighting coefficients

$$w_a = \frac{1}{(2.0)^2} = 0.25, \quad (68)$$

$$w_s = \frac{1}{(0.020)^2} = 2500, \quad (69)$$

and

$$w_F = \frac{1}{(500)^2} = 4 \times 10^{-6}. \quad (70)$$

Although the numerical magnitudes of these weighting coefficients are very different, they should not be compared directly because the corresponding physical quantities have different units and characteristic magnitudes. The reference-based formulation effectively normalizes each performance quantity before it contributes to the cost function.

The selected parameter set does not impose hard physical limits on body acceleration, suspension travel, or actuator force. Instead, the reference values

determine their relative importance within the optimization problem. The resulting controller was therefore designated as the *Balanced* controller, since no single performance objective was intentionally given exclusive priority.

Using these reference values, the matrices $\mathbf{Q}$, $\mathbf{N}$, and R defined in the previous section were constructed and the constant state-feedback gain $\mathbf{K}$ was obtained from the continuous-time LQR solution. The resulting actuator command is

$$F_a = -\mathbf{K}\mathbf{x}. \quad (71)$$

Because $\mathbf{K}$ remains constant throughout the simulation, the controller applies the same weighting of suspension objectives regardless of the road condition encountered by the vehicle. This controller therefore provides an active suspension response while retaining a **fixed control compromise**.

For the nominal quarter-car parameters, the resulting state-feedback gain was

$$\mathbf{K} = 10^3 \begin{bmatrix} 3.5830 & 1.4144 & 1.5477 & 0.3273 \end{bmatrix}. \quad (72)$$

3.4 Fixed-LQR Response on the Common Benchmark Road

The Balanced fixed-LQR controller was evaluated on the same 80 m composite benchmark road used for the passive suspension. Therefore, the mechanical parameters, road geometry, vehicle speed, and initial conditions remained unchanged, allowing the effect of the active controller to be evaluated directly.

The fixed controller was operated with an actuator-force magnitude limit of

$$|F_a| \leq 750 \text{ N}, \quad (73)$$

and a force-rate limit of

$$|\dot{F}_a| \leq 50\,000 \text{ N/s}. \quad (74)$$

The global response of the fixed-LQR controller is compared with the passive baseline in Table 4.

Table 4: Global performance comparison between the passive suspension and the Balanced fixed-LQR controller.

Model	Accel. RMS (m/s ²)	Accel. Peak (m/s ²)	Susp. Travel Peak (mm)	Tire Def. Peak (mm)	Force RMS (N)	Force Peak (N)
Passive	1.416	6.455	29.620	14.717	0	0
Fixed LQR	1.232	5.976	33.487	16.622	118.600	529.650

Relative to the passive suspension, the fixed-LQR controller reduced the global body-acceleration RMS by approximately 13.0% and the peak body acceleration by approximately 7.4%. This demonstrates that the introduction of

active control can improve the vertical body response without modifying the mechanical suspension parameters.

However, this improvement was accompanied by an increase of approximately 13.1% in peak suspension travel and 12.9% in peak tire deflection. The fixed controller therefore does not produce a uniform improvement in every performance metric. Instead, the active force redistributes the suspension response according to the compromise defined by the LQR cost function.

The response also remained strongly dependent on the type of road excitation. The segment-wise results of the fixed-LQR controller are given in Table 5.

Table 5: Segment-wise performance of the Balanced fixed-LQR controller.

Road Segment	Accel. RMS (m/s ²)	Accel. Peak (m/s ²)	Susp. Travel Peak (mm)	Tire Def. Peak (mm)	Force RMS (N)	Force Peak (N)
Mild Smooth 1	0.082	0.124	1.691	0.245	18.521	28.892
Long-Wave Undulation	0.620	0.928	12.687	1.708	140.550	216.740
Medium-Frequency	1.211	2.198	18.378	4.565	104.520	254.120
High-Frequency Rough	1.937	2.869	11.377	9.397	158.690	237.700
Bump Section	1.387	5.976	33.487	16.622	119.970	529.650
Wheel-Hop Region	2.206	3.336	11.769	12.280	181.980	274.910
Mild Smooth 2	0.478	2.484	11.492	11.521	39.094	182.840

The effect of the controller varies considerably between road segments. Large reductions in body acceleration are obtained during the mild and long-wave sections, whereas the improvement becomes smaller in the high-frequency and wheel-hop regions. The localized bump also produces the largest suspension travel and actuator-force peak.

This highlights a key limitation of the fixed-LQR approach. Although active control provides greater authority than the passive suspension, the same gain matrix $\mathbf{K}$ is applied under every disturbance condition. For example, during the bump, limiting suspension motion may become more important than preserving the same comfort-oriented compromise used on smoother road sections.

This motivates the use of multiple LQR controllers with different performance priorities, allowing the control compromise to change according to the dominant suspension response and naturally leading to the development of Model 3.

4 Model-3A, Gain-Scheduled LQR Control

4.1 Controller Bank and Model-3A Concept

The results of the fixed-LQR model showed that the preferred control objective may change with the character of the road disturbance. Model 3 was therefore developed by extending the single fixed-LQR controller into a bank of LQR controllers with different performance priorities.

All controllers were generated using the same quarter-car plant and the same generalized LQR formulation introduced in Section 3.2. Therefore, the

controller structure remained unchanged, while only the reference performance values defining the cost function were modified.

The resulting controller bank is summarized in Table 6.

Table 6: Reference values used for the LQR controller bank.

Controller	a_{ref} (m/s ²)	s_{ref} (m)	F_{ref} (N)
Balanced	2.0	0.020	500
Comfort	1.2	0.030	800
Low Force	2.0	0.020	350
Protection	3.0	0.012	300

Each controller represents a different compromise between the performance objectives. The *Balanced* controller corresponds to the fixed-LQR controller introduced previously. The *Comfort* controller applies a stronger relative penalty to body acceleration and allows greater suspension motion and actuator usage in order to improve ride comfort.

The *Low Force* controller retains the same acceleration and suspension-travel references as the Balanced controller but applies a larger penalty to actuator force. It therefore represents a lower-effort operating mode.

The *Protection* controller places greater emphasis on limiting suspension travel while accepting a reduced priority on body acceleration. It is intended for more demanding disturbances where mechanical response becomes the dominant concern.

Model-3A was introduced as a manually scheduled reference model to investigate whether changing the active-control priorities according to the disturbance condition could improve the overall suspension response.

In this model, the current road section is assumed to be known in advance. A controller is therefore selected directly according to the vehicle position on the benchmark road. The road profile itself is not estimated from the vehicle response; instead, the scheduling decision is prescribed *manually* for each known road segment.

The manually prescribed controller schedule is illustrated in Fig. 3. Low Force control is used during relatively mild sections, Comfort is assigned to body-dominated and localized bump disturbances, and Protection is selected for the high-frequency and wheel-hop regions.

4.2 Model 3A Performance on the Common Benchmark Road

Model 3A was evaluated on the same composite benchmark road used for the passive and fixed-LQR models. Since the road sections are assumed to be known

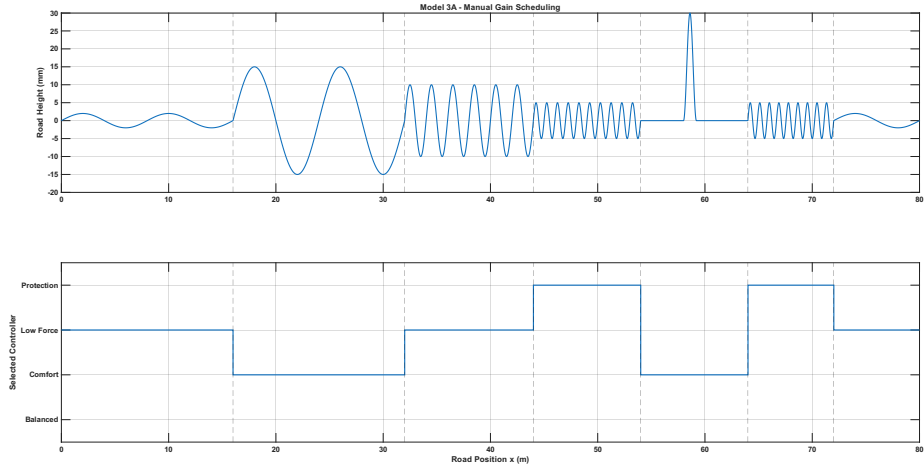


Figure 3: Common benchmark road and manually prescribed LQR controller schedule used in Model 3A.

in advance, the corresponding LQR controller is selected directly according to the prescribed schedule introduced in the previous subsection.

The global performance of Model 3A is compared with the Balanced fixed-LQR controller in Table 7.

Table 7: Global performance comparison between the fixed-LQR controller and Model 3A.

Model	Accel. RMS (m/s ²)	Accel. Peak (m/s ²)	Susp. Travel Peak (mm)	Tire Def. Peak (mm)	Force RMS (N)	Force Peak (N)
Fixed LQR	1.232	5.976	33.487	16.622	118.600	529.650
Model 3A	1.182	3.972	40.444	19.325	172.540	1450.000

The scheduled controller reduced the global body-acceleration RMS by approximately 4.0% and the peak body acceleration by approximately 33.5% relative to the fixed-LQR controller. The reduction in peak acceleration is particularly significant and demonstrates that changing the control priorities according to the disturbance condition can substantially alter the transient response.

However, the improvement in body acceleration was accompanied by increases in suspension travel, tire deflection, and actuator demand. The peak suspension travel increased to approximately 40.4 mm, while the maximum actuator force reached approximately 1450 N. Consequently, Model 3A should not be interpreted as a uniformly superior controller, but rather as a demonstration that gain scheduling can redistribute the suspension trade-off according to the selected performance objective.

The localized bump provides a clear example of this behavior. During this section, the Comfort controller reduces the peak body acceleration to approxi-

mately 3.97 m/s², compared with 5.98 m/s² for the fixed Balanced controller. This improvement is achieved at the expense of substantially larger suspension travel and actuator effort, with the actuator-force peak reaching approximately 1450 N.

These results confirm that changing the LQR objective can provide different advantages under different disturbance conditions. However, Model 3A relies on prior knowledge of the road section and does not consider practical actuator limitations when selecting the controller. These limitations motivate the development of Model 3B, in which the control mode is selected from measured vehicle responses rather than from known road position, while practical actuator constraints are also introduced.

5 Model-3B, Response-Based Gain-Scheduled LQR Control

Model 3A demonstrated that changing the LQR control objective according to the disturbance condition can modify the suspension trade-off. However, its controller schedule required prior knowledge of the road section. Such information would not generally be available to a vehicle without an external road-preview or road-classification system.

Model 3B was therefore developed to remove this dependency. Instead of selecting the controller according to road position, the scheduling decision is based entirely on measured vehicle responses. The same LQR controller bank introduced for Model 3A is retained, but the active controller is now selected online according to the observed dynamic condition of the suspension.

5.1 Model-3B Concept and Measured Responses

The purpose of the response-based scheduler is not to identify the road profile directly. Instead, it determines which type of suspension response is currently dominant and selects the corresponding LQR controller.

Four vehicle-response quantities are used by the scheduling logic: the sprung-mass acceleration,

$$\ddot{z}_s, \tag{75}$$

the unsprung-mass acceleration,

$$\ddot{z}_u, \tag{76}$$

the suspension travel,

$$z_s - z_u, \tag{77}$$

and the relative suspension velocity,

$$\dot{z}_s - \dot{z}_u. \quad (78)$$

These quantities provide complementary information about the current dynamic condition of the quarter-car system. Sprung-mass acceleration primarily describes body motion, while unsprung-mass acceleration provides an indication of wheel activity. Suspension travel indicates the relative motion between the two masses, and relative velocity is used to detect rapid transient events.

Table 8: Control modes used by the response-based scheduler.

Controller	Scheduling role
Low Force	Calm response and reduced actuator demand
Comfort	Body-dominated response with low wheel activity
Balanced	Mixed or ambiguous suspension response
Protection	Sustained wheel activity or rapid transient response

A key difference from Model 3A is therefore that no road-position information is required for the scheduling decision. Two different road profiles may produce similar vehicle responses and consequently lead to the same controller selection. Conversely, different dynamic responses within the same nominal road condition may result in different control modes.

Model 3B therefore performs *response-based gain scheduling* rather than direct road classification.

5.2 Scheduler Logic and Transient Detection

The measured responses cannot be used reliably from a single instantaneous sample alone, since the scheduler must distinguish sustained dynamic behavior from short transient events. Model 3B therefore uses two different time windows: a normal RMS window for general response classification and a shorter RMS window for rapid event detection.

The scheduler operates with a sampling period of

$$T_s = 0.002 \text{ s}, \quad (79)$$

corresponding to a sampling frequency of

$$f_s = 500 \text{ Hz}. \quad (80)$$

For a measured response y , the causal RMS quantity at sample k is evaluated as

$$y_{\text{RMS}}(k) = \sqrt{\frac{1}{N_w} \sum_{i=k-N_w+1}^k y_i^2}, \quad (81)$$

where N_w is the number of samples contained in the selected time window. For normal operating-condition classification, a window length of

$$T_w = 0.25 \text{ s} \quad (82)$$

is used, corresponding to

$$N_w = \frac{0.25}{0.002} = 125 \quad (83)$$

samples. RMS values of body acceleration, wheel acceleration, and suspension travel are calculated over this window.

A shorter wheel-acceleration RMS window of

$$T_{w,\text{fast}} = 0.05 \text{ s} \quad (84)$$

or 25 samples is additionally used for rapid transient detection. This fast quantity is used only to initiate a transient response and does not replace the normal RMS quantities used for sustained classification.

The principal thresholds used by the normal response classifier are summarized in Table 9.

Table 9: Main response thresholds used by the Model 3B scheduler.

Quantity	Threshold	Purpose
Body acceleration RMS	0.30 m/s ²	Calm/body-response classification
Suspension travel RMS	3 mm	Calm-condition classification
Wheel acceleration RMS	15 m/s ²	Maximum wheel activity for Low Force
Wheel acceleration RMS	3 m/s ²	Maximum wheel activity for Comfort
Suspension travel RMS	4 mm	Elevated travel for Comfort
Wheel acceleration RMS	20 m/s ²	Sustained Protection entry

The controller is selected according to a hierarchical decision structure. The *Low Force* controller is activated only when body acceleration, suspension travel, and wheel activity are all sufficiently low:

$$\begin{aligned} a_{\text{body,RMS}} &< 0.30 \text{ m/s}^2, \\ s_{\text{RMS}} &< 0.003 \text{ m}, \\ a_{\text{wheel,RMS}} &< 15 \text{ m/s}^2. \end{aligned} \quad (85)$$

The *Comfort* controller is selected when wheel activity remains low,

$$a_{\text{wheel,RMS}} < 3 \text{ m/s}^2, \quad (86)$$

while either the body acceleration or suspension travel indicates a more significant body-dominated response:

$$a_{\text{body,RMS}} \geq 0.30 \text{ m/s}^2 \quad (87)$$

or

$$s_{\text{RMS}} \geq 0.004 \text{ m.} \quad (88)$$

If the sustained wheel-acceleration RMS exceeds

$$a_{\text{wheel,RMS}} > 20 \text{ m/s}^2, \quad (89)$$

the *Protection* controller is selected.

Responses that do not clearly satisfy any of these conditions are classified as mixed or ambiguous, and the *Balanced* controller is used. Balanced control is also retained during the initial period before a complete normal RMS window becomes available.

The normal 0.25 s RMS window is suitable for identifying sustained response characteristics but may react too slowly to short disturbances such as a localized bump. A separate transient-detection mechanism was therefore introduced.

A transient sequence can be initiated when either the fast wheel-acceleration RMS exceeds

$$a_{\text{wheel,RMS,fast}} > 15 \text{ m/s}^2, \quad (90)$$

or the magnitude of the measured relative suspension velocity exceeds

$$|\dot{z}_s - \dot{z}_u| > 0.88 \text{ m/s.} \quad (91)$$

Once a transient is detected, the Protection controller is applied for

$$T_{\text{Protection}} = 60 \text{ ms.} \quad (92)$$

During this Protection interval, an additional confirmation condition is used to distinguish a significant shock event from other rapid wheel-dominated disturbances. A shock is confirmed if

$$|\dot{z}_s - \dot{z}_u| > 0.80 \text{ m/s} \quad (93)$$

for three consecutive new sensor samples. At the 500 Hz sampling rate, this corresponds to a confirmation duration of

$$3T_s = 6 \text{ ms.} \quad (94)$$

Following a confirmed shock, the scheduler switches from Protection to the Comfort controller for a 0.40 s follow-up period. The 0.40 s Comfort hold period was introduced as a tuning parameter to provide sufficient time for the post-shock body response to decay without excessive controller switching. This allows the initial rapid transient to be handled with the Protection-oriented controller while the subsequent body motion is treated with a more comfort-oriented control objective.

The resulting scheduler therefore combines two time scales. The slower RMS-based classifier determines the appropriate controller during sustained operation, while the fast event detector temporarily overrides the normal classification when a rapid transient is observed. This prevents the scheduling strategy from relying exclusively on global or slowly varying RMS measures.

The complete hierarchical scheduling logic is summarized in Fig. 4, where the normal RMS-based classification and the transient override mechanism are shown separately.

This hierarchical structure allows Model 3B to retain a stable response-based operating mode during sustained conditions while temporarily overriding the normal scheduler when a rapid transient is detected.

5.3 Actuator and Sensor Constraints

Unlike the manually scheduled reference model, Model 3B was evaluated with additional actuator and sensor constraints in order to represent a more realistic control implementation. These constraints do not modify the LQR gain matrices themselves, but limit how the commanded control force is physically applied and how quickly the scheduler receives new vehicle-response information.

5.3.1 Actuator Constraints

The actuator force was limited in magnitude according to

$$|F_a| \leq 750 \text{ N.} \quad (95)$$

In addition, the actuator was subjected to a force-rate limit of

$$|\dot{F}_a| \leq 50\,000 \text{ N/s.} \quad (96)$$

Since the controller operates with a sampling period of

$$T_s = 0.002 \text{ s,} \quad (97)$$

the maximum permitted change in actuator force between two consecutive samples is

$$\Delta F_{a,\max} = |\dot{F}_a|_{\max} T_s = 100 \text{ N.} \quad (98)$$

At each control update, the requested LQR force is first limited to the allowed force magnitude. The actuator then moves toward this target by no more than the permitted force increment. The applied force is therefore prevented from changing instantaneously, even when the scheduler switches between controllers.

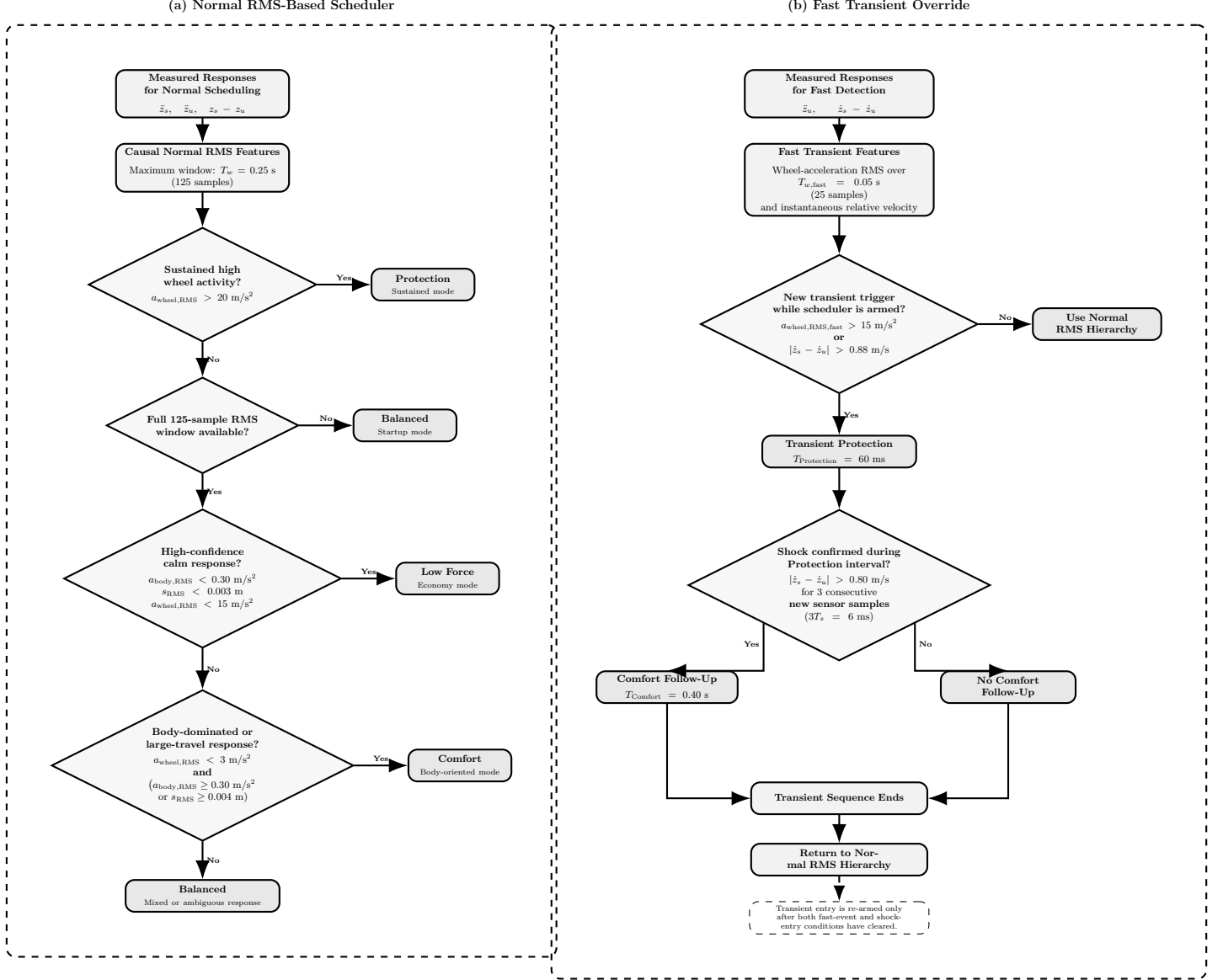


Figure 4: Hierarchical scheduling logic of the response-based gain-scheduled LQR controller used in Model 3B. The normal RMS-based hierarchy determines the controller during sustained operation, while the fast transient layer temporarily overrides this decision during rapid disturbances. For clarity, actuator limitations and sensor-timing effects are not shown in the scheduling diagram.

5.3.2 Sensor Timing

The response-based scheduler operates using measurements sampled at

$$f_s = 500 \text{ Hz}, \quad (99)$$

corresponding to a measurement interval of

$$T_s = 2 \text{ ms}. \quad (100)$$

A sensor transport delay of

$$t_{\text{delay}} = 4 \text{ ms} \quad (101)$$

was included in the nominal Model 3B implementation. The scheduling decision is therefore based on delayed measured responses rather than instantaneous access to the true plant states.

Between measurement updates, the most recently available sensor value is retained using a zero-order hold. Consequently, the scheduler uses only information that would have been available at the corresponding sampling instant.

For the nominal benchmark comparison, measurement noise was set to zero so that the influence of the scheduling logic and implementation constraints could be examined independently. Measurement noise is considered separately in the robustness analysis.

Importantly, sensor delay and measurement noise affect only the signals available to the scheduler. The reported performance metrics are calculated from the true simulated vehicle response rather than from the delayed or noisy measurements.

Table 10: Implementation constraints used in the nominal Model 3B simulation.

Parameter	Value
Maximum actuator force	750 N
Maximum actuator force rate	50 000 N/s
Sampling frequency	500 Hz
Sampling period	2 ms
Sensor transport delay	4 ms

These constraints distinguish Model 3B from the idealized scheduling concept of Model 3A and provide a more realistic basis for evaluating the response-based controller on the common benchmark road.

5.4 Model-3B Performance on the Common Benchmark Road

Model 3B was evaluated on the same 80 m composite benchmark road used for the previous suspension models. The mechanical parameters, vehicle speed, road geometry, and initial conditions were retained unchanged. The primary comparison was made against the Balanced fixed-LQR controller under the same actuator-force and force-rate limits.

Table 11: Global performance comparison between the Balanced fixed-LQR controller and Model 3B.

Model	Accel. RMS (m/s ²)	Accel. Peak (m/s ²)	Susp. Travel Peak (mm)	Tire Def. Peak (mm)	Force RMS (N)	Force Peak (N)
Fixed Balanced LQR	1.232	5.976	33.487	16.622	118.600	529.650
Model 3B	1.219	5.232	30.714	15.288	128.500	750.000

Relative to the fixed-LQR controller, Model 3B reduced the global body acceleration RMS by approximately 1.0%, the peak body acceleration by 12.4%, the peak suspension travel by 8.3%, and the peak tire deflection by 8.0%. These improvements were accompanied by an approximately 8.3% increase in actuator-force RMS, while the peak force reached the imposed 750 N limit.

The relatively small change in acceleration RMS is consistent with the purpose of the scheduler, which is not to minimize a single global metric but to vary the control compromise according to the observed vehicle response.

Figure 5 compares the controller selections of Models 3A and 3B over the common benchmark road.

Unlike Model 3A, whose controller sequence is prescribed according to road position, Model 3B selects the controller from the measured vehicle response. Multiple controller transitions may therefore occur within a single road segment.

The overall distribution of controller usage is shown in Figure 6.

The Balanced, Comfort, Low Force, and Protection controllers accounted for approximately 24.1%, 24.9%, 25.8%, and 25.2% of the simulation time, respectively. This distributed usage confirms that the scheduler does not remain concentrated around a single control objective.

The localized bump produced the largest transient response, with a peak body acceleration of approximately 5.23 m/s², a suspension-travel peak of 30.7 mm, and a tire-deflection peak of 15.3 mm. The actuator reached the imposed 750 N force limit during this event. In the wheel-hop region, the acceleration RMS increased to approximately 2.24 m/s² while the suspension travel remained near 10.8 mm, illustrating that the dominant performance requirement varies with the vehicle response.

Compared with the manually scheduled Model 3A, Model 3B substantially reduces the required actuator effort and mechanical excursions. The peak suspension travel decreases from approximately 40.4 mm to 30.7 mm, while the actuator-force peak is limited from 1450 N to 750 N. This reflects the transition

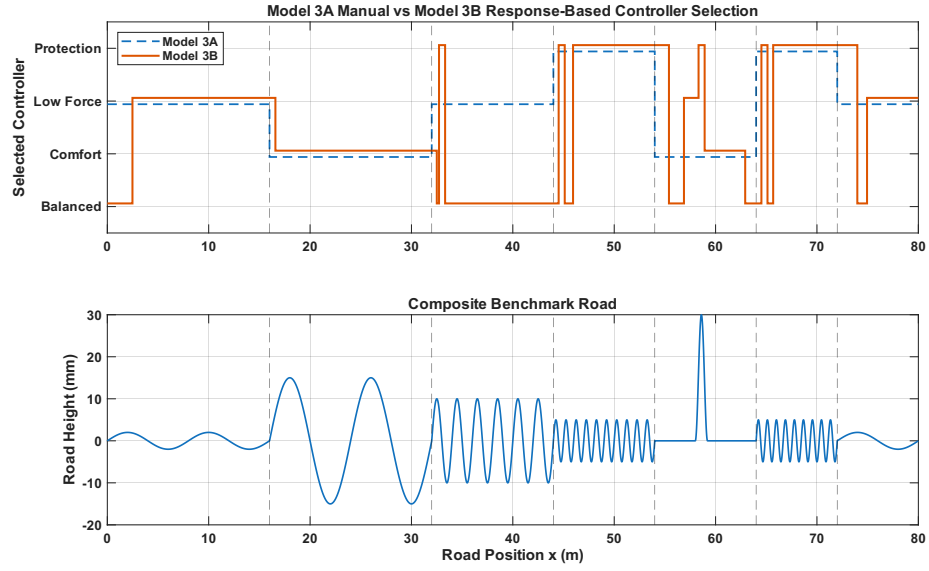


Figure 5: Controller-selection behavior of Models 3A and 3B on the composite benchmark road. The upper panel compares the selected controllers, while the lower panel shows the corresponding road profile.

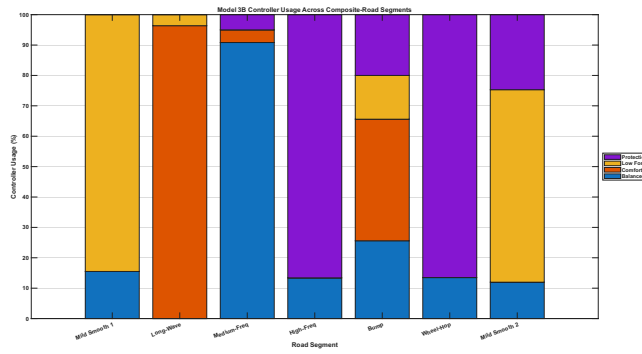


Figure 6: Controller usage of Model 3B over the complete composite benchmark road.

from an idealized manually scheduled reference to a response-based controller operating under explicit implementation constraints.

6 Robustness and Realism Evaluation

The nominal composite-road results demonstrate the behavior of Model 3B under the conditions used during controller development. Additional tests were therefore performed to examine whether the same fixed controller bank, scheduler thresholds, time windows, and hold periods remain effective under changes in disturbance severity, vehicle speed, and road profile.

No controller gains or scheduler parameters were retuned during the following tests.

6.1 Bump Height and Vehicle-Speed Sensitivity

The first robustness study examines the response of Model 3B to isolated bumps with different heights and vehicle speeds. Three vehicle speeds,

$$v = \{5, 10, 15\} \text{ m/s}, \quad (102)$$

and three bump heights,

$$h_b = \{15, 30, 45\} \text{ mm}, \quad (103)$$

were considered, producing nine test conditions in total. The bump length was fixed at 1.2 m. No controller gains, scheduling thresholds, time windows, or hold periods were retuned between the simulations.

Table 12: Model 3B response to variations in bump height and vehicle speed.

Speed (m/s)	Bump Height (mm)	Accel. Peak (m/s ²)	Travel Peak (mm)	Tire Def. Peak (mm)	Force Peak (N)	Protection Time (ms)	Comfort Time (ms)
5	15	1.206	14.334	2.365	443.53	0	128
5	30	2.565	24.707	4.731	339.61	0	158
5	45	4.063	37.034	7.051	504.32	60	214
10	15	3.361	16.708	8.226	268.87	60	0
10	30	5.239	30.803	15.277	750.00	60	400
10	45	8.146	46.171	23.005	750.00	60	400
15	15	4.303	15.996	12.841	386.64	60	400
15	30	8.579	31.512	25.598	750.00	60	400
15	45	12.359	47.229	38.428	750.00	60	400

As expected, increasing bump height generally increased the peak body acceleration, suspension travel, and tire deflection. Vehicle speed also had a strong influence on the transient severity, particularly on body acceleration and tire deformation.

The actuator-force limit became active for the larger disturbances. At 10 and 15 m/s, the 30 and 45 mm bumps reached the imposed 750 N force limit, while the lower-speed cases remained below saturation.

Whenever the transient Protection mode was activated, its duration remained 60 ms, consistent with the fixed Protection hold time defined in the scheduler. The time spent in Comfort varied between cases, reflecting both the transient-follow-up logic and the normal response-based controller selection within the evaluation interval.

6.2 Performance on Unseen Mixed Roads

To evaluate whether the scheduling logic was excessively dependent on the nominal composite benchmark road, Model 3B was also tested on three previously unseen mixed road profiles: Mild, Medium, and Rough. Each profile was evaluated at vehicle speeds of 5, 10, and 15 m/s.

The same controller gains, scheduling thresholds, RMS windows, transient logic, and actuator limits were retained. Model 3B was compared with the fixed Balanced LQR controller under the same physical constraints. Negative percentage changes indicate reductions relative to the fixed controller.

Each unseen road was constructed as a deterministic three-component multi-sine profile,

$$z_{r,\text{raw}}(x) = \sum_{i=1}^3 A_i \sin\left(\frac{2\pi x}{\lambda_i} + \phi_i\right), \quad (104)$$

where A_i , λ_i , and ϕ_i denote the amplitude, wavelength, and phase of each spatial component, respectively. The parameters used for the three profiles are given in Table 13.

Table 13: Parameters of the unseen mixed-road profiles.

Profile	Amplitudes A_i (mm)	Wavelengths λ_i (m)	Phases ϕ_i (rad)
Mixed Mild	[1.5, 1.0, 0.5]	[7.0, 3.1, 1.4]	[0.15, 0.90, 1.70]
Mixed Medium	[3.0, 2.0, 1.0]	[6.2, 2.4, 1.1]	[0.35, 1.20, 2.10]
Mixed Rough	[5.0, 3.0, 1.5]	[5.4, 1.8, 0.9]	[0.55, 1.45, 2.40]

A smooth spatial envelope was applied to avoid artificial discontinuities at the beginning and end of the disturbance. Each 60 m test road was zero before $x = 5$ m, smoothly ramped to full amplitude over the next 5 m, remained at full amplitude from $x = 10$ to 50 m, and was smoothly reduced to zero between $x = 50$ and 55 m. Performance metrics were evaluated only over the full-amplitude region from $x = 10$ to 50 m.

The results show that Model 3B does not uniformly outperform the fixed Balanced controller on every metric. Instead, the scheduler changes the control compromise according to the measured vehicle response.

On the Mild profiles, the scheduler primarily favored the Low Force controller. This substantially reduced actuator-force RMS, although in some cases it increased body acceleration and mechanical motion.

Table 14: Performance change of Model 3B relative to the fixed Balanced controller on unseen mixed roads.

Road Profile	Speed (m/s)	Δ Accel. RMS (%)	Δ Travel Peak (%)	Δ Tire Peak (%)	Δ Force RMS (%)
Mixed Mild	5	19.09	5.93	15.24	-27.58
Mixed Medium	5	14.03	6.85	10.37	-29.19
Mixed Rough	5	-20.66	17.58	12.63	83.88
Mixed Mild	10	13.75	10.15	8.04	-28.04
Mixed Medium	10	0.08	0.00	0.00	0.03
Mixed Rough	10	2.34	1.87	-0.49	-2.67
Mixed Mild	15	3.72	-2.90	-4.10	-31.28
Mixed Medium	15	0.04	0.00	0.00	-0.08
Mixed Rough	15	8.73	7.88	0.70	-21.21

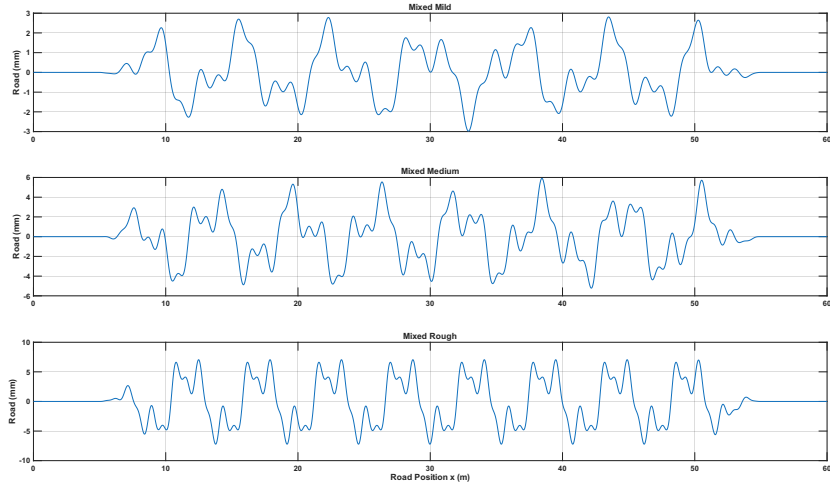


Figure 7: Unseen mixed-road profiles used for robustness evaluation. All three profiles were generated independently of the nominal composite benchmark road.

A different behavior was observed on the Rough profiles. At 5 m/s, Comfort dominated the response and reduced acceleration RMS by approximately 20.7%, at the cost of increased actuator effort, suspension travel, and tire deflection. At higher rough-road speeds, the scheduler shifted toward Balanced and Protection modes.

For the Medium profile at 10 and 15 m/s, the scheduler remained almost entirely in the Balanced mode, producing performance nearly identical to the fixed-LQR reference. This is consistent with the intended role of Balanced as the default controller for mixed or ambiguous response conditions.

The unseen-road results therefore indicate that the scheduler can operate on disturbance profiles outside the nominal composite benchmark without requiring retuning. However, its benefit should be interpreted as condition-dependent redistribution of the suspension trade-off rather than uniform improvement of every performance metric.

6.3 Combined Sensor and Implementation Realism

The final robustness study evaluates the complete Model 3B implementation with the actuator limits, sensor timing, shock-confirmation logic, and measurement noise applied simultaneously. No controller gains, scheduler thresholds, time windows, or hold periods were retuned.

For each road scenario, the results were compared with the same locked Model 3B controller using zero sensor delay and ideal measurements. Therefore, the percentage changes reported in this section represent the combined influence of the implemented sensor timing and measurement realism, rather than a comparison with the fixed-LQR controller.

Table 15: Effect of combined sensor timing and measurement realism on Model 3B.

Scenario	Noise	Mode Mismatch (%)	Δ Accel. RMS (%)	Δ Accel. Peak (%)	Δ Travel Peak (%)	Δ Force RMS (%)
Nominal Composite	Ideal	0.87	0.18	0.09	0.07	-0.82
Nominal Composite	High	1.17	0.17	0.10	0.08	-0.91
Mixed Medium, 10 m/s	Ideal	0.10	0.03	0.00	0.00	-0.03
Mixed Medium, 10 m/s	High	0.15	0.05	0.00	0.00	-0.05
Mixed Rough, 15 m/s	Ideal	7.80	-0.14	0.36	0.06	0.53
Mixed Rough, 15 m/s	High	7.28	-0.14	0.36	0.15	0.57

For the nominal composite road, even the High-noise condition produced only a 1.17% controller-mode mismatch relative to the zero-delay reference. Changes in the global performance metrics remained below approximately 1%.

The Mixed Medium case was even less sensitive to the added measurement realism, with a mode mismatch below 0.2% and essentially unchanged vehicle-response metrics.

The largest mode mismatch occurred for the Mixed Rough profile at 15 m/s, where approximately 7–8% of the controller selections differed from the zero-delay reference. Despite this larger scheduling difference, the resulting changes

in acceleration, suspension travel, and actuator-force RMS remained below approximately 1%. This indicates that small differences in the exact switching sequence do not necessarily produce large changes in the overall vehicle response.

Table 16: Nominal-event validation under measurement noise.

Noise Level	Bump Confirmed	Confirmation Position (m)	Wheel-Hop Confirmed
Ideal	Yes	58.52	No
Low	Yes	58.52	No
Moderate	Yes	58.52	No
High	Yes	58.58	No

The nominal bump was correctly confirmed as a shock under every tested noise level. The confirmation position remained nearly unchanged, shifting only from approximately 58.52 m to 58.58 m under the High-noise condition. In contrast, the wheel-hop region was not incorrectly confirmed as a shock in any of the tested cases.

This result is particularly important for the transient scheduler, since it indicates that the shock-confirmation logic remains selective under the tested measurement-noise conditions rather than responding indiscriminately to high wheel activity.

Overall, the robustness studies indicate that Model 3B retains its intended scheduling behavior under variations in disturbance severity, vehicle speed, unseen road profiles, sensor timing, and measurement noise without scenario-specific retuning. The results do not constitute a formal safety guarantee, but they demonstrate that the proposed response-based scheduling strategy remains numerically stable and functionally consistent across the tested operating conditions.

7 Conclusion

This study investigated the progressive development of an active quarter-car suspension controller, beginning with a passive suspension and extending to a response-based gain-scheduled LQR architecture. The passive model established the fundamental compromise between ride comfort, suspension travel, and road holding. A fixed active LQR controller subsequently demonstrated that actuator authority can improve the suspension response, but also showed that a single gain matrix represents only one fixed compromise across different disturbance conditions.

Model 3A introduced a bank of LQR controllers with different performance priorities and demonstrated the potential benefit of changing the control objective according to the disturbance condition. However, this model required prior knowledge of the road position and produced relatively high actuator and

suspension demands. Model 3B therefore replaced the predefined road schedule with a hierarchical response-based scheduler using measured body acceleration, wheel acceleration, suspension travel, and relative suspension velocity. Fast transient detection was combined with slower RMS-based classification, while actuator-force, force-rate, sensor-sampling, and transport-delay constraints were also included.

On the common benchmark road, Model 3B reduced the peak body acceleration by approximately 12.4%, the peak suspension travel by 8.3%, and the peak tire deflection by 8.0% relative to the fixed Balanced LQR controller. The body-acceleration RMS changed by only approximately 1%, while actuator-force RMS increased by approximately 8.3%. These results demonstrate that the main benefit of the proposed scheduler is not the uniform minimization of a single performance quantity, but the ability to modify the suspension trade-off according to the dominant measured response.

Additional robustness studies were performed without retuning the controller gains, scheduling thresholds, RMS windows, or transient hold times. The scheduler remained operational across variations in bump height and vehicle speed and produced distinct controller-selection behavior on previously unseen mixed-road profiles. The final combined realism tests further showed that sensor delay and measurement noise caused only small changes in the overall vehicle-response metrics. The nominal bump was consistently identified as a confirmed shock under all tested noise conditions, while the wheel-hop region was not incorrectly classified as a shock.

Overall, the results support the feasibility of response-based gain-scheduled LQR control as a method for adapting active-suspension priorities without requiring direct knowledge of the road profile. Nevertheless, the proposed controller remains a proof-of-concept based on a linear quarter-car model. The transient hold periods are tuned parameters rather than formally optimized safety conditions, and explicit mechanical safety limits were not imposed. Future work could therefore investigate state-dependent transitions between Protection and Comfort modes, nonlinear suspension and actuator models, explicit constraint handling, and extension of the control architecture to higher-order vehicle models.

References

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